

Customer Churn Prediction & LTV Engine

Week 3 • Day 7

Week 3 Review & API Documentation

Complete API documentation, review all Week 3 work, and prepare for dashboard week

■ Day Objectives

- ◆ Complete FastAPI documentation with detailed descriptions and examples
- ◆ Create API_DOCUMENTATION.md in the project
- ◆ Test the complete API end-to-end
- ◆ Ensure all 7 Week 3 commits are on GitHub
- ◆ Prepare for Week 4: dashboard and Docker

■ 1. Complete API Documentation

■ Create docs/API_DOCUMENTATION.md

```
# Customer Churn & LTV Prediction API
## Version 1.0.0 | Base URL: http://localhost:8000

## Endpoints

### GET /health
Returns API health status.
Response: {status, timestamp, models_loaded}

### POST /predict
```

```
Predicts churn probability for a single customer.
Request Body: CustomerInput (see schema)
Response: ChurnPredictionResponse
- churn_probability: float [0-1]
- churn_prediction: 'Will Churn' | 'Will Stay'
- risk_level: 'High' | 'Medium' | 'Low'
- predictive_ltv: float (USD)
- ltv_segment: 'High Value' | 'Medium Value' | 'Low Value'
- top_risk_factors: list of SHAP explanations
- recommendation: retention action string
```

```
### POST /predict/batch
```

Processes a list of CustomerInput objects.

Response: {total_customers, high_risk_count, predictions[]}

```
### POST /predict/csv
```

Accepts CSV file upload, returns predictions CSV.

```
## Authentication
```

Currently no authentication (development mode).

Production: add API key header: X-API-Key: <key>

```
## Error Codes
```

- 200: Success
- 422: Validation error (invalid input)
- 500: Internal server error (model failure)

■ 2. Week 3 Review Checklist

- Day 15: LTV calculated for all customers, customers_ltv table in DB
- Day 16: LTV regression model trained (MAE < \$200)
- Day 17: FastAPI skeleton running, /health endpoint works
- Day 18: POST /predict returns full prediction response
- Day 19: POST /predict/batch working
- Day 20: 5 API tests pass
- Day 21: API documentation complete, 7 commits on GitHub

■ 3. Commit & Week 3 Wrap-up

■ Week3-Day7 Commit

```
git add .
git commit -m "Week3-Day7: API documentation complete, Week 3 review, all tests passing"
git push origin main
```

■ Estimated Time: 4 – 5 hours

■ **Tip:** Week 3 complete! You now have a fully working ML API. Week 4 builds the dashboard, containerises everything with Docker, and produces your final deliverables.

